
Game-based Learning: Motivation Activation and Memory Reinforcement Teaching Innovation

Wang Yanan
BJZGCA
maxwang0809@163.com

Abstract

This study focuses on professional course instruction for adolescents, exploring the impact of game-based instructional design grounded in psychological mechanisms on students' learning interest and long-term memory retention. Integrating the Hook behavioral model and the Fear of Missing Out (FOMO) effect, the research constructs a "Motivation-Memory Dual-Drive" game-based learning framework. Quasi-experiments were conducted in real classroom settings with fourth-grade and sixth-grade students. The experimental group used a self-designed gamified learning system featuring mechanisms such as daily login rewards, time-limited challenges, and an integrated points-level-badge system, while the control group received traditional instruction. This study provides theoretical support and practical case studies for the design of game-based learning within smart education contexts.

Keywords: Gamification Incentive Mechanisms; Hook Model; FOMO; Learning Motivation; Interactive Learning

1 Introduction

In today's era of information technology sweeping through the educational landscape, traditional teaching models are facing unprecedented challenges. Issues such as shortened student attention spans, insufficient learning motivation, and rapid knowledge forgetting are becoming increasingly prominent, especially in language disciplines that rely heavily on memorization. Taking middle school English vocabulary learning as an example, numerous studies indicate that students generally find repetitive rote memorization tedious and boring, lacking intrinsic drive. This leads to low learning efficiency and even creates a vicious cycle of "learning quickly, forgetting even faster". Although teachers continuously attempt to strengthen memory through dictation and recitation, these methods rely heavily on external supervision and short-term stimulation, making it difficult to cultivate sustainable learning habits. Therefore, how to stimulate students' interest, enhance their willingness to participate actively, and scientifically improve long-term knowledge retention has become a core issue solved in current educational reform.

In recent years, "Gamification in Education" [1] has emerged as a novel teaching strategy, gradually attracting attention from both academia and practitioners. Gamification refers to the application of game design elements (such as points, badges, leaderboards, and task levels) to non-game contexts to enhance user engagement and behavioral stickiness. A large body of empirical research confirms that well-designed gamification can effectively improve student motivation, task completion rates, and classroom participation [2]. However, many existing gamification practices in education remain superficial, merely introducing reward mechanisms without considering the underlying deep psychological drivers. This often results in "learning for the sake of playing" rather than "playing for the sake of learning," ultimately failing to achieve genuine cognitive engagement and lasting behavioral change [3].

In this context, relevant theories from behavioral psychology and cognitive science offer important insights for deepening gamified learning design. On one hand, Nir Eyal's Hook Model reveals the four-stage path of habit formation: Trigger, Action, Variable Reward, and Investment. This provides a theoretical framework for constructing a closed-loop of sustainable learning behaviors [4]. While this model has been widely applied in digital product design, it has recently been gradually transferred to educational settings to promote autonomous learning behaviors. On the other hand, FOMO (Fear of Missing Out), a social psychological phenomenon prevalent among adolescents, can be leveraged through time-limited challenges, real-time rankings, and peer progress notifications to stimulate students' sense of urgency and competitive awareness, thereby increasing participation frequency [5].

However, there is currently a lack of research that systematically integrates these various psychological mechanisms to build a comprehensive instructional design that balances "motivation activation" with "memory reinforcement." Most studies either prioritize interest at the expense of cognitive laws or emphasize review efficiency while neglecting emotional experience, failing to achieve the organic unity of being "interesting" and "effective". Based on this, the present study focuses on the context of teaching. It attempts to construct a gamified learning framework that integrates the Hook Model to drive learning habits, the FOMO mechanism to enhance participation motivation. Through instructional design and preliminary practice, this study explores its feasibility and application value. The research aims to answer: How can the synergistic effects of psychological principles be utilized to design an instructional intervention that not only attracts sustained student participation but also promotes deep knowledge retention? This study not only contributes to enriching the theoretical system of gamified learning but also provides a practical reference for curriculum innovation and technology platform development in smart education environments.

2 Related work

Definition and Development of Gamification in Education Gamification in Education refers to the systematic application of game design elements (such as points, badges, leaderboards, task levels, and immediate feedback) to non-game educational contexts to enhance learners' motivation, engagement, and behavioral stickiness [6]. Its core is not to turn teaching into a game, but to leverage game mechanics to stimulate intrinsic or extrinsic motivation and promote goal-oriented learning behaviors [7]. Since its emergence around 2010, gamification in education has rapidly become a research hotspot in the field of educational technology. Early practices were mostly concentrated in language learning, programming education, and corporate training [8]. With the popularity of mobile internet and smart terminals, gamification has gradually been integrated into K-12 classrooms, showing potential in repetitive tasks such as vocabulary memorization and mathematics practice [9]. Meta-analyses have further confirmed that well-designed gamification interventions can significantly improve students' learning motivation, task perseverance, and academic performance [10]. However, some scholars have pointed out that if reward mechanisms are mechanically applied without considering cognitive load and alignment with learning objectives, gamification may produce a counterproductive effect of being "superficially lively but substantively inefficient" [11].

The Application Potential of the Hook Model in Education Proposed by Nir Eyal, the Hook Model explains how users form usage habits through four stages: Trigger, Action, Variable Reward, and Investment. Initially applied to consumer digital product design (such as social media and short video platforms), this model has been gradually introduced into the field of education in recent years to construct a closed-loop of sustainable autonomous learning behaviors. In an educational context, "Triggers" can come from task reminders pushed by teachers or system notifications; "Action" refers to students completing a small but achievable learning task (e.g., mastering a unit of knowledge); "Variable Rewards" provide unpredictable positive feedback—enhancing dopamine-driven engagement—through random badges, surprise points, or peer likes; and the "Investment" phase encourages students to increase sunk costs through personalization, content creation, or progress accumulation, thereby enhancing long-term retention willingness [12]. Preliminary studies indicate that learning applications designed based on the Hook Model can effectively improve students' daily active rates and task completion rates. However, its systematic integration into formal classroom teaching is still in the exploratory stage.

3 Theoretical Framework Construction

3.1 Motivation-Interaction Dual-Driven Gamified Learning Model

Current gamified learning practices often fall into two major pitfalls: first, an over-reliance on surface-level elements like badges and leaderboards while neglecting the underlying psychological mechanisms that drive sustained engagement ("prioritizing form over mechanism"); second, a disconnect between gamified activities and deep learning objectives and cognitive processes ("separation of motivation and cognition"), leading to students "playing without learning."

To address these issues, this study constructs the "Motivation-Interaction Dual-Driven Gamified Learning Model" (MID-GLM). The model posits that merely stimulating motivation without guiding effective interaction leads to shallow participation, while well-designed interactions lacking sustained motivational drivers struggle to initiate and maintain learners' active exploration. Therefore, the motivation layer and the interaction layer must function as dual engines, working synergistically to guide learners through a complete cycle: from "capturing attention" to "facilitating memory and understanding," and finally to "forming stable habits," thereby achieving efficient and lasting learning outcomes.

The model features a dual-loop structure: the outer **motivation layer** serves as the external engine, responsible for dynamically sustaining learning drive. It integrates the internalized application of the Hook Model and the positive guidance of the Fear of Missing Out (FOMO) mechanism to trigger, maintain, and reinforce learners' willingness to participate. For example, external triggers (e.g., push notifications) and internal triggers (e.g., curiosity) guide learners into the context, while variable rewards (e.g., random incentives) sustain interest, ultimately encouraging investment and forming a continuous cycle. Simultaneously, the FOMO mechanism, through time-limited challenges and progress visualization, stimulates learners' social comparison tendencies, motivating them to participate actively to avoid falling behind.

The inner **interaction layer** serves as the internal core, focusing on constructing a closed loop of effective learning experiences to ensure that learners, once attracted by motivation, engage in substantive cognitive processing and knowledge construction. This layer emphasizes:

- **Clear Task Structure:** Decomposing learning content into progressively challenging tasks aligned with the "zone of proximal development" theory to avoid frustration.
- **Timely and Multidimensional Feedback Mechanisms:** Including immediate feedback, formative feedback, and diverse feedback forms (system, social, intrinsic) to help learners continuously adjust their strategies.
- **Embedded Social Support:** Transforming learning into a social construction process through collaborative tasks or knowledge-sharing communities to enhance a sense of belonging.

3.2 Motivation Layer: Synergistic Drive of the Hook Model and FOMO

The motivation layer aims to address the question of "why students are willing to participate continuously." By integrating the Hook Model and the FOMO mechanism, it constructs stable behavioral inertia.

Hook Model: Systematizes learning behavior into a cyclically reinforcing process:

- **Trigger:** Guides learners into the context through various triggers (e.g., smart push notifications, teacher reminders).
- **Action:** Designs low-threshold "minimum viable action" tasks to reduce the burden of initiation.
- **Variable Reward:** Provides unpredictable positive feedback, activating the brain's reward system and enhancing behavioral stickiness.
- **Investment:** Encourages learners to invest resources in the system, creating sunk costs that lay the groundwork for subsequent triggers, ultimately forming a self-reinforcing habit loop.

3.3 Interaction Layer: Core Mechanisms Supporting Deep Learning

The interaction layer aims to address the question of "how to ensure students truly learn while participating." It deeply integrates game elements with cognitive principles through four core mechanisms.

- **Task Design Mechanism:** Based on the Ebbinghaus forgetting curve, constructs spaced repetition task flows to optimize memory retention. Tasks adopt diverse formats and embed game narrative elements like "level unlocking" to give learning activities a sense of story and purpose. Difficulty increases spirally, aligning with the "zone of proximal development" theory.
- **Feedback System Mechanism:** Moves beyond simple right/wrong judgments to provide immediate,
- **Social Incentive Mechanism:** Transforms learning from an isolated activity into a social construction process.
- **Progress Visualization Mechanism:** Employs tools like "learning journey maps," "skill tree lighting," and "weekly achievement walls" to make abstract learning processes tangible, enhancing learners' sense of purpose and control. This not only complements the reward system of the motivation layer but also supports learners' metacognitive development, enabling them to autonomously monitor progress, adjust strategies, and derive intrinsic satisfaction.

4 Experiment

4.1 Results analysis

Statistics	Baseline 1	Test 1	Baseline 2	Test 2
Number	33	33	33	33
Mean	76.36	78.97	85.93	84.52
Median	80	82	86.7	85
Standard deviation	18.34	14.19	11.21	9.67
Range	60	49.5	43.3	35
1/4 percentile	65	68.5	80	78.5

This analysis is based on the test scores of 33 subjects across four assessments. Specifically, Test 1 and Test 3 represent the pre- and post-intervention results for software multiple-choice question (MCQ) detection, while Test 2 and Test 4 represent the pre- and post-intervention results for classroom written examinations. The analysis examines the data through the dimensions of central tendency, dispersion, and quantile distribution.

Central Tendency From the perspective of central tendency, both assessment types show significant performance improvements, with a slightly higher improvement magnitude observed in the software MCQ detection.

Dispersion (Variability) In terms of dispersion, the variability in scores decreased significantly for both types of assessments as the testing progressed, indicating a narrowing of the performance gap among students and increasingly balanced overall results.

Quantile Distribution The quantile distribution further corroborates the aforementioned trends, highlighting significant improvements in lower scores while higher scores remained stable.

The comparative analysis reflects a significant positive impact of the learning process on student performance. The continuous reduction in the interquartile range indicates that low-performing students improved markedly, while high-performing students maintained stable results. This resulted in a more concentrated and stable score distribution, demonstrating the effectiveness of the intervention in promoting balanced development across the student cohort.

This analysis is based on the test scores of 26 subjects across four distinct stages (Baseline 1, Test 1, Baseline 2, Test 2), examining the data through the dimensions of central tendency, dispersion, and quantile distribution. From the perspective of central tendency, the average student performance shows a trend of initial increase followed by stabilization. In terms of dispersion, the data generally

Statistics	Baseline 1	Test 1	Baseline 2	Test 2
Number	26	26	26	26
Mean	0.7844	0.82	0.845	0.8969
Median	0.805	0.854	0.82	0.93
Standard deviation	0.1555	0.1420	0.1324	0.1210
Range	0.60	0.55	0.55	0.45
1/4 percentile	0.75	0.82	0.78	0.85

indicates a gradual narrowing of score gaps and an increase in stability. The quantile distribution provides a more granular perspective.

5 Conclusion

Based on the Motivation-Interaction Dual-Driven Gamified Learning Model (MID-GLM), this study employs a quasi-experimental design and multi-source data analysis to validate the positive effects and applicability boundaries of gamified learning systems on the learning behaviors and academic performance of fourth and sixth-grade primary school students. (1) **The effectiveness of the mechanism has been confirmed:** students' post-test scores in both software multiple-choice questions and classroom written exams showed significant improvement, with a continuous reduction in score dispersion. This indicates that the intervention not only enhanced overall performance but also promoted balanced development among the student population. (2) **The dual-driven mechanism operates synergistically:** positive feedback from over 90% of students further confirmed that the integration of the motivation layer (incorporating the Hook Model and FOMO mechanisms) and the interaction layer (through spaced repetition tasks, multi-dimensional feedback, and social incentives) transformed learning from a passive task into a purposeful "learning game."

While the model demonstrates clear advantages in scenarios such as foundational knowledge acquisition, learning initiation, and memorization, it has not yet been tested for learning objectives requiring higher-order thinking or complex creativity. Caution is warranted regarding the risks of "taskification" and superficial engagement. The study also revealed that some students (approximately 10%) developed a reliance on external reward mechanisms, highlighting the importance and challenges of transitioning from external incentives to internal motivation. Future design and research should focus on **verifying the long-term benefits of the mechanism** (e.g., by extending experimental durations), **promoting motivation internalization** (e.g., replacing simple point systems with "knowledge mastery maps"), **enhancing learner autonomy** (by offering path choices and challenge customization), and **exploring dynamic adjustment mechanisms tailored to different learners and subject characteristics**. This will facilitate a deeper integration of motivation stimulation and cognitive science, ultimately guiding learners to cultivate autonomous and profound learning habits.

References

- [1] Serafeim A. Triantafyllou, Christos Georgiadis, and Theodosios Sapounidis. Gamification in education and training: A literature review. *International Review of Education*, pages 1–35, 2025.
- [2] Taufik Ikhsan Slamet and Chen Meng. Gamification in collaborative learning: Synthesizing evidence through meta-analysis. *Journal of Computers in Education*, pages 1–37, 2025.
- [3] Bangkit Puri Nugroho and Emilia Fitriana Dewi. Enhancing performance management system through gamification: A comprehensive review and improvement. *INVEST: Jurnal Inovasi Bisnis dan Akuntansi*, 6(1):118–124, 2025.
- [4] Nir Eyal. *Hooked: How to build habit-forming products*. Penguin Uk, 2014.
- [5] Muhammad Nadeem, Melinda Oroszlanyova, and Wael Farag. Effect of digital game-based learning on student engagement and motivation. *Computers*, 12(9):177, 2023.

- [6] Sebastian Deterding et al. From game design elements to gamefulness: defining “gamification”. In *Proceedings of the 15th international academic MindTrek conference: Envisioning future media environments*, 2011.
- [7] Karl M. Kapp. *The gamification of learning and instruction fieldbook: Ideas into practice*. John Wiley & Sons, 2013.
- [8] Juho Hamari, Jonna Koivisto, and Harri Sarsa. Does gamification work?—a literature review of empirical studies on gamification. In *2014 47th Hawaii international conference on system sciences*. IEEE, 2014.
- [9] Michael Sailer and Lisa Homner. The gamification of learning: A meta-analysis. *Educational psychology review*, 32(1):77–112, 2020.
- [10] Richard N. Landers. Developing a theory of gamified learning: Linking serious games and gamification of learning. *Simulation & gaming*, 45(6):752–768, 2014.
- [11] Luis De-Marcos et al. An empirical study comparing gamification and social networking on e-learning. *Computers & education*, 75:82–91, 2014.
- [12] N. Eyal. *Hooked: How to build habit-forming products*. Portfolio, 2014.